Recursion Vs Iteration 1

Summation of different series

1. 1 + 2 + 3 + 4 + ... upto n'th term.

```
#include<stdio.h>
int N;

int getSum(int i)
{
    if (i > N) {
        printf("\n");
        return 0;
    }
    printf("%d ", i);
    return i + getSum(i+1);
}

int main()
{
    int i;
    int sum;

    scanf("%d", &N);

    printf("Iteration: \n");
    sum = 0;
    for (i=1; i<=N; i++) {
        printf("%d ", i);
        sum += i;
    }
    printf("\n");
    printf("Sum = %d\n", sum);

    printf("\nRecursion: \n");
    sum = getSum(1);
    printf("Sum = %d\n", sum);

    return 0;
}
```

2. 2 + 4 + 6 + 8 + ... upto n'th term.

```
#include<stdio.h>
int N;

int getSum(int i)
{
```

```
    if (i > N) {
        printf("\n");
        return 0;
    }
    printf("%d ", 2*i);
    return 2*i + getSum(i+1);
}

int main()
{
    int i;
    int sum;

    scanf("%d", &N);

    printf("Iteration: \n");
    sum = 0;
    for (i=1; i<=N; i++) {
        printf("%d ", 2*i);
        sum += 2*i;
    }
    printf("\n");
    printf("Sum = %d\n", sum);

    printf("\nRecursion: \n");
    sum = getSum(1);
    printf("Sum = %d\n", sum);

    return 0;
}
```

3. 1 + 3 + 5 + 7 + ... upto n'th term.

```
#include<stdio.h>
int N;

int getSum(int i)
{
    if (i > N) {
        printf("\n");
        return 0;
    }
    printf("%d ", (2*i-1));
    return (2*i-1) + getSum(i+1);
}

int main()
{
    int i;
    int sum;
```

```
    scanf("%d", &N);

    printf("Iteration: \n");
    sum = 0;
    for (i=1; i<=N; i++) {
        printf("%d ", (2*i-1));
        sum += (2*i-1);
    }
    printf("\n");
    printf("Sum = %d\n", sum);

    printf("\nRecursion: \n");
    sum = getSum(1);
    printf("Sum = %d\n", sum);

    return 0;
}
```

4. 4 + 12 + 20 + 28 + ... upto n'th term.

```
// 4*1 + 4*3 + 4*5 + 4*7 + ...........
#include<stdio.h>
int N;

int getSum(int i)
{
    if (i > N) {
        printf("\n");
        return 0;
    }
    printf("%d ", 4*(2*i-1));
    return 4*(2*i-1) + getSum(i+1);
}

int main()
{
    int i;
    int sum;

    scanf("%d", &N);

    printf("Iteration: \n");
    sum = 0;
    for (i=1; i<=N; i++) {
        printf("%d ", 4*(2*i-1));
        sum += 4*(2*i-1);
    }
    printf("\n");
    printf("Sum = %d\n", sum);
```

```
    printf("\nRecursion: \n");
    sum = getSum(1);
    printf("Sum = %d\n", sum);

    return 0;
}
```

5. 2 + 5 + 8 + 11 + ... upto n'th term.

```
// 2 = 3*1 - 1
// 5 = 3*2 - 1
// 8 = 3*3 - 1
// 11 = 3*4 - 1
#include<stdio.h>
int N;

int getSum(int i)
{
    if (i > N) {
        printf("\n");
        return 0;
    }
    printf("%d ", 3*i-1);
    return 3*i-1 + getSum(i+1);
}

int main()
{
    int i;
    int sum;

    scanf("%d", &N);

    printf("Iteration: \n");
    sum = 0;
    for (i=1; i<=N; i++) {
        printf("%d ", 3*i-1);
        sum += 3*i-1;
    }
    printf("\n");
    printf("Sum = %d\n", sum);

    printf("\nRecursion: \n");
    sum = getSum(1);
    printf("Sum = %d\n", sum);

    return 0;
}
```

6. 1.2 + 2.3 + 3.4 + ... upto n'th term.

```
// 1.2 = 1.(1+1)
// 2.3 = 2.(2+1)
// 3.4 = 3.(3+1)
#include<stdio.h>
int N;

int getSum(int i)
{
    if (i > N) {
        printf("\n");
        return 0;
    }
    printf("%d ", i*(i+1));
    return i*(i+1) + getSum(i+1);
}

int main()
{
    int i;
    int sum;

    scanf("%d", &N);

    printf("Iteration: \n");
    sum = 0;
    for (i=1; i<=N; i++) {
        printf("%d ", i*(i+1));
        sum += i*(i+1);
    }
    printf("\n");
    printf("Sum = %d\n", sum);

    printf("\nRecursion: \n");
    sum = getSum(1);
    printf("Sum = %d\n", sum);

    return 0;
}
```

7. 2.1 + 5.3 + 8.5 + ... upto n'th term.

```
// 2.1 = (3*1-1)*(2*1-1)
// 5.3 = (3*2-1)*(2*2-1)
// 8.5 = (3*3-1)*(2*3-1)
#include<stdio.h>
int N;

int getSum(int i)
```

```
{
    if (i > N) {
        printf("\n");
        return 0;
    }
    printf("%d ", (3*i-1)*(2*i-1));
    return (3*i-1)*(2*i-1) + getSum(i+1);
}

int main()
{
    int i;
    int sum;

    scanf("%d", &N);

    printf("Iteration: \n");
    sum = 0;
    for (i=1; i<=N; i++) {
        printf("%d ", (3*i-1)*(2*i-1));
        sum += (3*i-1)*(2*i-1);
    }
    printf("\n");
    printf("Sum = %d\n", sum);

    printf("\nRecursion: \n");
    sum = getSum(1);
    printf("Sum = %d\n", sum);

    return 0;
}
```

8. 1.3 + 3.5 + 5.7 + ... upto n'th term.

```
// 1.3 = (2*1-1)*(2*1+1)
// 3.5 = (2*2-1)*(2*2+1)
// 5.7 = (2*3-1)*(2*3+1)
#include<stdio.h>
int N;

int getSum(int i)
{
    if (i > N) {
        printf("\n");
        return 0;
    }
    printf("%d ", (2*i-1)*(2*i+1));
    return (2*i-1)*(2*i+1) + getSum(i+1);
}
```

```
int main()
{
    int i;
    int sum;

    scanf("%d", &N);

    printf("Iteration: \n");
    sum = 0;
    for (i=1; i<=N; i++) {
        printf("%d ", (2*i-1)*(2*i+1));
        sum += (2*i-1)*(2*i+1);
    }
    printf("\n");
    printf("Sum = %d\n", sum);

    printf("\nRecursion: \n");
    sum = getSum(1);
    printf("Sum = %d\n", sum);

    return 0;
}
```

9. 1^2 + 3^2 + 5^2 + ... upto n'th term.

```
// 1^2 = (2*1-1)*(2*1-1)
// 3^2 = (2*2-1)*(2*2-1)
// 5^2 = (2*3-1)*(2*3-1)
#include<stdio.h>
int N;

int getSum(int i)
{
    if (i > N) {
        printf("\n");
        return 0;
    }
    printf("%d ", (2*i-1)*(2*i-1));
    return (2*i-1)*(2*i-1) + getSum(i+1);
}

int main()
{
    int i;
    int sum;

    scanf("%d", &N);

    printf("Iteration: \n");
    sum = 0;
```

```
    for (i=1; i<=N; i++) {
        printf("%d ", (2*i-1)*(2*i-1));
        sum += (2*i-1)*(2*i-1);
    }
    printf("\n");
    printf("Sum = %d\n", sum);

    printf("\nRecursion: \n");
    sum = getSum(1);
    printf("Sum = %d\n", sum);

    return 0;
}
```

10. 1.1^2 + 2.3^2 + 3.5^2 + ... upto n'th term.

```
// 1.1^2 = 1*(2*1-1)*(2*1-1)
// 2.3^2 = 2*(2*2-1)*(2*2-1)
// 3.5^2 = 3*(2*3-1)*(2*3-1)
#include<stdio.h>
int N;

int getSum(int i)
{
    if (i > N) {
        printf("\n");
        return 0;
    }
    printf("%d ", i*(2*i-1)*(2*i-1));
    return i*(2*i-1)*(2*i-1) + getSum(i+1);
}

int main()
{
    int i;
    int sum;

    scanf("%d", &N);

    printf("Iteration: \n");
    sum = 0;
    for (i=1; i<=N; i++) {
        printf("%d ", i*(2*i-1)*(2*i-1));
        sum += i*(2*i-1)*(2*i-1);
    }
    printf("\n");
    printf("Sum = %d\n", sum);

    printf("\nRecursion: \n");
    sum = getSum(1);
```

```
    printf("Sum = %d\n", sum);

    return 0;
}
```

11. 1.2^2 + 2.3^2 + 3.4^2 + ... upto n'th term.

```
// 1.2^2 = 1*(1+1)*(1+1)
// 2.3^2 = 2*(2+1)*(2+1)
// 3.4^2 = 3*(3+1)*(3+1)
#include<stdio.h>
int N;

int getSum(int i)
{
    if (i > N) {
        printf("\n");
        return 0;
    }
    printf("%d ", i*(i+1)*(i+1));
    return i*(i+1)*(i+1) + getSum(i+1);
}

int main()
{
    int i;
    int sum;

    scanf("%d", &N);

    printf("Iteration: \n");
    sum = 0;
    for (i=1; i<=N; i++) {
        printf("%d ", i*(i+1)*(i+1));
        sum += i*(i+1)*(i+1);
    }
    printf("\n");
    printf("Sum = %d\n", sum);

    printf("\nRecursion: \n");
    sum = getSum(1);
    printf("Sum = %d\n", sum);

    return 0;
}
```

12. 1^2.2^2 + 2^2.3^2 + 3^2.4^2 + ... upto n'th term.

```
// 1^2.2^2 = 1*1*(1+1)*(1+1)
```

```
// 2^2.3^2 = 2*2*(2+1)*(2+1)
// 3^2.4^2 = 3*3*(3+1)*(3+1)
// i^2.(i+1)^2 = i*i*(i+1)*(i+1)
#include<stdio.h>
int N;

int getSum(int i)
{
    if (i > N) {
        printf("\n");
        return 0;
    }
    printf("%d ", i*i*(i+1)*(i+1));
    return i*i*(i+1)*(i+1) + getSum(i+1);
}

int main()
{
    int i;
    int sum;

    scanf("%d", &N);

    printf("Iteration: \n");
    sum = 0;
    for (i=1; i<=N; i++) {
        printf("%d ", i*i*(i+1)*(i+1));
        sum += i*i*(i+1)*(i+1);
    }
    printf("\n");
    printf("Sum = %d\n", sum);

    printf("\nRecursion: \n");
    sum = getSum(1);
    printf("Sum = %d\n", sum);

    return 0;
}
```

13. 1.2.3 + 2.3.4 + 3.4.5 + ... upto n'th term.

```
// 1.2.3 = 1*(1+1)*(1+2)
// 2.3.4 = 2*(2+1)*(2+2)
// 3.4.5 = 3*(3+1)*(3+2)
// i*(i+1)*(i+2)
#include<stdio.h>
int N;

int getSum(int i)
{
```

```
    if (i > N) {
        printf("\n");
        return 0;
    }
    printf("%d ", i*(i+1)*(i+2));
    return i*(i+1)*(i+2) + getSum(i+1);
}

int main()
{
    int i;
    int sum;

    scanf("%d", &N);

    printf("Iteration: \n");
    sum = 0;
    for (i=1; i<=N; i++) {
        printf("%d ", i*(i+1)*(i+2));
        sum += i*(i+1)*(i+2);
    }
    printf("\n");
    printf("Sum = %d\n", sum);

    printf("\nRecursion: \n");
    sum = getSum(1);
    printf("Sum = %d\n", sum);

    return 0;
}
```

14. 1.3.5.7 + 3.5.7.9 + 5.7.9.11 + ... upto n'th term.

```
// 1.3.5.7 = 105
// 3.5.7.9 = 945
// 5.7.9.11 = 3465
// (2*i-1)*(2*i+1)*(2*i+3)*(2*i+5)
#include<stdio.h>
int N;

int getSum(int i)
{
    if (i > N) {
        printf("\n");
        return 0;
    }
    printf("%d ", (2*i-1)*(2*i+1)*(2*i+3)*(2*i+5));
    return (2*i-1)*(2*i+1)*(2*i+3)*(2*i+5) + getSum(i+1);
}
```

```
int main()
{
    int i;
    int sum;

    scanf("%d", &N);

    printf("Iteration: \n");
    sum = 0;
    for (i=1; i<=N; i++) {
        printf("%d ", (2*i-1)*(2*i+1)*(2*i+3)*(2*i+5));
        sum += (2*i-1)*(2*i+1)*(2*i+3)*(2*i+5);
    }
    printf("\n");
    printf("Sum = %d\n", sum);

    printf("\nRecursion: \n");
    sum = getSum(1);
    printf("Sum = %d\n", sum);

    return 0;
}
```

15. 2.5.8 + 5.8.11 + 8.11.14 + ... upto n'th term.

```
// 2.5.8 = 80
// 5.8.11 = 440
// 8.11.14 = 1232
// (3*i-1)*(3*i+2)*(3*i+5)
#include<stdio.h>
int N;

int getSum(int i)
{
    if (i > N) {
        printf("\n");
        return 0;
    }
    printf("%d ", (3*i-1)*(3*i+2)*(3*i+5));
    return (3*i-1)*(3*i+2)*(3*i+5) + getSum(i+1);
}

int main()
{
    int i;
    int sum;

    scanf("%d", &N);

    printf("Iteration: \n");
```

```
    sum = 0;
    for (i=1; i<=N; i++) {
        printf("%d ", (3*i-1)*(3*i+2)*(3*i+5));
        sum += (3*i-1)*(3*i+2)*(3*i+5);
    }
    printf("\n");
    printf("Sum = %d\n", sum);

    printf("\nRecursion: \n");
    sum = getSum(1);
    printf("Sum = %d\n", sum);

    return 0;
}
```

16. 5.6.7 + 6.7.8 + 7.8.9 + ... upto n'th term.

```
// 5.6.7 = 210
// 6.7.8 = 336
// 7.8.9 = 504
// (i+4)*(i+5)*(i+6)
#include<stdio.h>
int N;

int getSum(int i)
{
    if (i > N) {
        printf("\n");
        return 0;
    }
    printf("%d ", (i+4)*(i+5)*(i+6));
    return (i+4)*(i+5)*(i+6) + getSum(i+1);
}

int main()
{
    int i;
    int sum;

    scanf("%d", &N);

    printf("Iteration: \n");
    sum = 0;
    for (i=1; i<=N; i++) {
        printf("%d ", (i+4)*(i+5)*(i+6));
        sum += (i+4)*(i+5)*(i+6);
    }
    printf("\n");
    printf("Sum = %d\n", sum);
```

```
    printf("\nRecursion: \n");
    sum = getSum(1);
    printf("Sum = %d\n", sum);

    return 0;
}
```

Recursion Vs Iteration 2

Pyramid

1.

```
1
1   2
1   2   3
1   2   3   4
1   2   3   4   5
```

```
#include<stdio.h>
int N;

void iteration()
{
    int i, j;
    for (i=1; i<=N; i++) {
        for (j=1; j<=i; j++)
            printf("%4d", j);
        printf("\n");
    }
}

void recursion2(int i, int j)
{
    if (j > i) {
        printf("\n");
        return;
    }
    printf("%4d", j);
    recursion2(i, j+1);
}

void recursion1(int i)
{
    if (i > N)
        return;
    recursion2(i, 1);
    recursion1(i+1);
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(1);
```

```
    return 0;
}
```

2.

```
1
2   2
3   3   3
4   4   4   4
5   5   5   5   5
```

```
#include<stdio.h>
int N;

void iteration()
{
    int i, j;
    for (i=1; i<=N; i++) {
        for (j=1; j<=i; j++)
            printf("%4d", i);
        printf("\n");
    }
}

void recursion2(int i, int j)
{
    if (j > i) {
        printf("\n");
        return;
    }
    printf("%4d", i);
    recursion2(i, j+1);
}

void recursion1(int i)
{
    if (i > N)
        return;
    recursion2(i, 1);
    recursion1(i+1);
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(1);
```

```
    return 0;
}
```

3.

```
1
0   0
1   1   1
0   0   0   0
1   1   1   1   1
```

```
#include<stdio.h>
int N;

void iteration()
{
    int i, j;
    for (i=1; i<=N; i++) {
        for (j=1; j<=i; j++)
            printf("%4d", i%2);
        printf("\n");
    }
}

void recursion2(int i, int j)
{
    if (j > i) {
        printf("\n");
        return;
    }
    printf("%4d", i%2);
    recursion2(i, j+1);
}

void recursion1(int i)
{
    if (i > N)
        return;
    recursion2(i, 1);
    recursion1(i+1);
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(1);
```

```
    return 0;
}
```

4.

```
1
1   0
1   0   1
1   0   1   0
1   0   1   0   1
```

```
#include<stdio.h>
int N;

void iteration()
{
    int i, j;
    for (i=1; i<=N; i++) {
        for (j=1; j<=i; j++)
            printf("%4d", j%2);
        printf("\n");
    }
}

void recursion2(int i, int j)
{
    if (j > i) {
        printf("\n");
        return;
    }
    printf("%4d", j%2);
    recursion2(i, j+1);
}

void recursion1(int i)
{
    if (i > N)
        return;
    recursion2(i, 1);
    recursion1(i+1);
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(1);
```

```
    return 0;
}
```

5.

```
2
3   4
4   5   6
5   6   7   8
6   7   8   9  10
```

```
#include<stdio.h>
int N;

void iteration()
{
    int i, j;
    for (i=1; i<=N; i++) {
        for (j=1; j<=i; j++)
            printf("%4d", i+j);
        printf("\n");
    }
}

void recursion2(int i, int j)
{
    if (j > i) {
        printf("\n");
        return;
    }
    printf("%4d", i+j);
    recursion2(i, j+1);
}

void recursion1(int i)
{
    if (i > N)
        return;
    recursion2(i, 1);
    recursion1(i+1);
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(1);
```

```
    return 0;
}
```

6.

```
1
2   3
3   4   5
4   5   6   7
5   6   7   8   9
```

```
#include<stdio.h>
int N;

void iteration()
{
    int i, j;
    for (i=1; i<=N; i++) {
        for (j=1; j<=i; j++)
            printf("%4d", i+j-1);
        printf("\n");
    }
}

void recursion2(int i, int j)
{
    if (j > i) {
        printf("\n");
        return;
    }
    printf("%4d", i+j-1);
    recursion2(i, j+1);
}

void recursion1(int i)
{
    if (i > N)
        return;
    recursion2(i, 1);
    recursion1(i+1);
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(1);
```

```
    return 0;
}
```

7.

```
0
1   0
0   1   0
1   0   1   0
0   1   0   1   0
```

```
#include<stdio.h>
int N;

void iteration()
{
    int i, j;
    for (i=1; i<=N; i++) {
        for (j=1; j<=i; j++)
            printf("%4d", (i+j)%2);
        printf("\n");
    }
}

void recursion2(int i, int j)
{
    if (j > i) {
        printf("\n");
        return;
    }
    printf("%4d", (i+j)%2);
    recursion2(i, j+1);
}

void recursion1(int i)
{
    if (i > N)
        return;
    recursion2(i, 1);
    recursion1(i+1);
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(1);
```

```
    return 0;
}
```

8.

```
1
0   1
1   0   1
0   1   0   1
1   0   1   0   1
```

```
#include<stdio.h>
int N;

void iteration()
{
    int i, j;
    for (i=1; i<=N; i++) {
        for (j=1; j<=i; j++)
            printf("%4d", (i+j-1)%2);
        printf("\n");
    }
}

void recursion2(int i, int j)
{
    if (j > i) {
        printf("\n");
        return;
    }
    printf("%4d", (i+j-1)%2);
    recursion2(i, j+1);
}

void recursion1(int i)
{
    if (i > N)
        return;
    recursion2(i, 1);
    recursion1(i+1);
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(1);
```

```
    return 0;
}
```

9.

```
A
B   B
C   C   C
D   D   D   D
E   E   E   E   E
```

```
#include<stdio.h>
int N;

void iteration()
{
    int i, j;
    for (i=1; i<=N; i++) {
        for (j=1; j<=i; j++)
            printf("%4c", i+64);
        printf("\n");
    }
}

void recursion2(int i, int j)
{
    if (j > i) {
        printf("\n");
        return;
    }
    printf("%4c", i+64);
    recursion2(i, j+1);
}

void recursion1(int i)
{
    if (i > N)
        return;
    recursion2(i, 1);
    recursion1(i+1);
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(1);
```

```
    return 0;
}
```

10.

```
A
A   B
A   B   C
A   B   C   D
A   B   C   D   E
```

```
#include<stdio.h>
int N;

void iteration()
{
    int i, j;
    for (i=1; i<=N; i++) {
        for (j=1; j<=i; j++)
            printf("%4c", j+64);
        printf("\n");
    }
}

void recursion2(int i, int j)
{
    if (j > i) {
        printf("\n");
        return;
    }
    printf("%4c", j+64);
    recursion2(i, j+1);
}

void recursion1(int i)
{
    if (i > N)
        return;
    recursion2(i, 1);
    recursion1(i+1);
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(1);
```

```
    return 0;
}
```

11.

```
1   2   3   4   5
1   2   3   4
1   2   3
1   2
1
```

```
#include<stdio.h>
int N;

void iteration()
{
    int i, j;
    for (i=N; i>=1; i--) {
        for (j=1; j<=i; j++)
            printf("%4d", j);
        printf("\n");
    }
}

void recursion2(int i, int j)
{
    if (j > i) {
        printf("\n");
        return;
    }
    printf("%4d", j);
    recursion2(i, j+1);
}

void recursion1(int i)
{
    if (i == 0)
        return;
    recursion2(i, 1);
    recursion1(i-1);
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(N);
```

```
    return 0;
}
```

12.

```
5   5   5   5   5
4   4   4   4
3   3   3
2   2
1
```

```
#include<stdio.h>
int N;

void iteration()
{
    int i, j;
    for (i=N; i>=1; i--) {
        for (j=1; j<=i; j++)
            printf("%4d", i);
        printf("\n");
    }
}

void recursion2(int i, int j)
{
    if (j > i) {
        printf("\n");
        return;
    }
    printf("%4d", i);
    recursion2(i, j+1);
}

void recursion1(int i)
{
    if (i == 0)
        return;
    recursion2(i, 1);
    recursion1(i-1);
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(N);
```

```
    return 0;
}
```

13.

```
1   1   1   1   1
0   0   0   0
1   1   1
0   0
1
```

```
#include<stdio.h>
int N;

void iteration()
{
    int i, j;
    for (i=N; i>=1; i--) {
        for (j=1; j<=i; j++)
            printf("%4d", i%2);
        printf("\n");
    }
}

void recursion2(int i, int j)
{
    if (j > i) {
        printf("\n");
        return;
    }
    printf("%4d", i%2);
    recursion2(i, j+1);
}

void recursion1(int i)
{
    if (i == 0)
        return;
    recursion2(i, 1);
    recursion1(i-1);
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(N);
```

```
    return 0;
}
```

14.

```
1   0   1   0   1
1   0   1   0
1   0   1
1   0
1
```

```
#include<stdio.h>
int N;

void iteration()
{
    int i, j;
    for (i=N; i>=1; i--) {
        for (j=1; j<=i; j++)
            printf("%4d", j%2);
        printf("\n");
    }
}

void recursion2(int i, int j)
{
    if (j > i) {
        printf("\n");
        return;
    }
    printf("%4d", j%2);
    recursion2(i, j+1);
}

void recursion1(int i)
{
    if (i == 0)
        return;
    recursion2(i, 1);
    recursion1(i-1);
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(N);
```

```
    return 0;
}
```

15.

```
6   7   8   9  10
5   6   7   8
4   5   6
3   4
2
```

```
#include<stdio.h>
int N;

void iteration()
{
    int i, j;
    for (i=N; i>=1; i--) {
        for (j=1; j<=i; j++)
            printf("%4d", i+j);
        printf("\n");
    }
}

void recursion2(int i, int j)
{
    if (j > i) {
        printf("\n");
        return;
    }
    printf("%4d", i+j);
    recursion2(i, j+1);
}

void recursion1(int i)
{
    if (i == 0)
        return;
    recursion2(i, 1);
    recursion1(i-1);
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(N);
```

```
    return 0;
}
```

16.

```
5   6   7   8   9
4   5   6   7
3   4   5
2   3
1
```

```
#include<stdio.h>
int N;

void iteration()
{
    int i, j;
    for (i=N; i>=1; i--) {
        for (j=1; j<=i; j++)
            printf("%4d", i+j-1);
        printf("\n");
    }
}

void recursion2(int i, int j)
{
    if (j > i) {
        printf("\n");
        return;
    }
    printf("%4d", i+j-1);
    recursion2(i, j+1);
}

void recursion1(int i)
{
    if (i == 0)
        return;
    recursion2(i, 1);
    recursion1(i-1);
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(N);
```

```
    return 0;
}
```

17.

```
0   1   0   1   0
1   0   1   0
0   1   0
1   0
0
```

```
#include<stdio.h>
int N;

void iteration()
{
    int i, j;
    for (i=N; i>=1; i--) {
        for (j=1; j<=i; j++)
            printf("%4d", (i+j)%2);
        printf("\n");
    }
}

void recursion2(int i, int j)
{
    if (j > i) {
        printf("\n");
        return;
    }
    printf("%4d", (i+j)%2);
    recursion2(i, j+1);
}

void recursion1(int i)
{
    if (i == 0)
        return;
    recursion2(i, 1);
    recursion1(i-1);
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(N);
```

```
    return 0;
}
```

18.

```
1   0   1   0   1
0   1   0   1
1   0   1
0   1
1
```

```
#include<stdio.h>
int N;

void iteration()
{
    int i, j;
    for (i=N; i>=1; i--) {
        for (j=1; j<=i; j++)
            printf("%4d", (i+j-1)%2);
        printf("\n");
    }
}

void recursion2(int i, int j)
{
    if (j > i) {
        printf("\n");
        return;
    }
    printf("%4d", (i+j-1)%2);
    recursion2(i, j+1);
}

void recursion1(int i)
{
    if (i == 0)
        return;
    recursion2(i, 1);
    recursion1(i-1);
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(N);
```

```
    return 0;
}
```

19.

```
E   E   E   E   E
D   D   D   D
C   C   C
B   B
A
```

```
#include<stdio.h>
int N;

void iteration()
{
    int i, j;
    for (i=N; i>=1; i--) {
        for (j=1; j<=i; j++)
            printf("%4c", i+64);
        printf("\n");
    }
}

void recursion2(int i, int j)
{
    if (j > i) {
        printf("\n");
        return;
    }
    printf("%4c", i+64);
    recursion2(i, j+1);
}

void recursion1(int i)
{
    if (i == 0)
        return;
    recursion2(i, 1);
    recursion1(i-1);
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(N);
```

```
    return 0;
}
```

20.

```
A   B   C   D   E
A   B   C   D
A   B   C
A   B
A
```

```
#include<stdio.h>
int N;

void iteration()
{
    int i, j;
    for (i=N; i>=1; i--) {
        for (j=1; j<=i; j++)
            printf("%4c", j+64);
        printf("\n");
    }
}

void recursion2(int i, int j)
{
    if (j > i) {
        printf("\n");
        return;
    }
    printf("%4c", j+64);
    recursion2(i, j+1);
}

void recursion1(int i)
{
    if (i == 0)
        return;
    recursion2(i, 1);
    recursion1(i-1);
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(N);
```

```
    return 0;
}
```

21.

```
                1
            1   2
        1   2   3
    1   2   3   4
1   2   3   4   5
```

```
#include<stdio.h>
int N;

void iteration()
{
    int i, j;
    for (i=1; i<=N; i++) {
        for (j=1; j<=N-i; j++)
            printf("    ");
        for (j=1; j<=i; j++)
            printf("%4d", j);
        printf("\n");
    }
}

void recursion2(int i, int j)
{
    if (j > i) {
        printf("\n");
        return;
    }
    printf("%4d", j);
    recursion2(i, j+1);
}

void recursion3(int i, int j)
{
    if (j > N-i)
        return;
    printf("    ");
    recursion3(i, j+1);
}

void recursion1(int i)
{
    if (i > N)
        return;
    recursion3(i, 1);
```

```
    recursion2(i, 1);
    recursion1(i+1);
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(1);
    return 0;
}
```

22.

```
               1
           2   2
       3   3   3
   4   4   4   4
5  5   5   5   5
```

```
#include<stdio.h>
int N;

void iteration()
{
    int i, j;
    for (i=1; i<=N; i++) {
        for (j=1; j<=N-i; j++)
            printf("    ");
        for (j=1; j<=i; j++)
            printf("%4d", i);
        printf("\n");
    }
}

void recursion2(int i, int j)
{
    if (j > i) {
        printf("\n");
        return;
    }
    printf("%4d", i);
    recursion2(i, j+1);
}

void recursion3(int i, int j)
{
```

```
    if (j > N-i)
        return;
    printf("    ");
    recursion3(i, j+1);
}

void recursion1(int i)
{
    if (i > N)
        return;
    recursion3(i, 1);
    recursion2(i, 1);
    recursion1(i+1);
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(1);
    return 0;
}
```

23.

```
                1
            0   0
        1   1   1
    0   0   0   0
1   1   1   1   1
```

```
#include<stdio.h>
int N;

void iteration()
{
    int i, j;
    for (i=1; i<=N; i++) {
        for (j=1; j<=N-i; j++)
            printf("    ");
        for (j=1; j<=i; j++)
            printf("%4d", i%2);
        printf("\n");
    }
}

void recursion2(int i, int j)
```

```
{
    if (j > i) {
        printf("\n");
        return;
    }
    printf("%4d", i%2);
    recursion2(i, j+1);
}

void recursion3(int i, int j)
{
    if (j > N-i)
        return;
    printf("    ");
    recursion3(i, j+1);
}

void recursion1(int i)
{
    if (i > N)
        return;
    recursion3(i, 1);
    recursion2(i, 1);
    recursion1(i+1);
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(1);
    return 0;
}
```

24.

```
               1
           1   0
       1   0   1
   1   0   1   0
1  0   1   0   1
```

```
#include<stdio.h>
int N;

void iteration()
{
```

```
    int i, j;
    for (i=1; i<=N; i++) {
        for (j=1; j<=N-i; j++)
            printf("    ");
        for (j=1; j<=i; j++)
            printf("%4d", j%2);
        printf("\n");
    }
}

void recursion2(int i, int j)
{
    if (j > i) {
        printf("\n");
        return;
    }
    printf("%4d", j%2);
    recursion2(i, j+1);
}

void recursion3(int i, int j)
{
    if (j > N-i)
        return;
    printf("    ");
    recursion3(i, j+1);
}

void recursion1(int i)
{
    if (i > N)
        return;
    recursion3(i, 1);
    recursion2(i, 1);
    recursion1(i+1);
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(1);
    return 0;
}
```

25.

```
        1
```

```
      2   2
    3   3   3
  4   4   4   4
5   5   5   5   5
```

```
#include<stdio.h>
int N;

void iteration()
{
    int i, j;
    for (i=1; i<=N; i++) {
        for (j=1; j<=N-i; j++)
            printf("  ");
        for (j=1; j<=i; j++)
            printf("%4d", i);
        printf("\n");
    }
}

void recursion2(int i, int j)
{
    if (j > i) {
        printf("\n");
        return;
    }
    printf("%4d", i);
    recursion2(i, j+1);
}

void recursion3(int i, int j)
{
    if (j > N-i)
        return;
    printf("  ");
    recursion3(i, j+1);
}

void recursion1(int i)
{
    if (i > N)
        return;
    recursion3(i, 1);
    recursion2(i, 1);
    recursion1(i+1);
}

int main()
{
    scanf("%d", &N);
```

```
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(1);
    return 0;
}
```

26.

```
        A
      B   B
    C   C   C
  D   D   D   D
E   E   E   E   E
```

```
#include<stdio.h>
int N;

void iteration()
{
    int i, j;
    for (i=1; i<=N; i++) {
        for (j=1; j<=N-i; j++)
            printf("  ");
        for (j=1; j<=i; j++)
            printf("%4c", i+64);
        printf("\n");
    }
}

void recursion2(int i, int j)
{
    if (j > i) {
        printf("\n");
        return;
    }
    printf("%4c", i+64);
    recursion2(i, j+1);
}

void recursion3(int i, int j)
{
    if (j > N-i)
        return;
    printf("  ");
    recursion3(i, j+1);
}

void recursion1(int i)
```

```
{
    if (i > N)
        return;
    recursion3(i, 1);
    recursion2(i, 1);
    recursion1(i+1);
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(1);
    return 0;
}
```

27.

```
        A
      B   B
    C   C   C
  D   D   D   D
E   E   E   E   E
  D   D   D   D
    C   C   C
      B   B
        A
```

```
#include<stdio.h>
int N;

void iteration()
{
    int i, j;
    for (i=1; i<=N; i++) {
        for (j=1; j<=N-i; j++)
            printf("  ");
        for (j=1; j<=i; j++)
            printf("%4c", i+64);
        printf("\n");
    }
    for (i=N-1; i>=1; i--) {
        for (j=1; j<=N-i; j++)
            printf("  ");
        for (j=1; j<=i; j++)
            printf("%4c", i+64);
        printf("\n");
```

```
    }
}

void recursion2(int i, int j)
{
    if (j > i) {
        printf("\n");
        return;
    }
    printf("%4c", i+64);
    recursion2(i, j+1);
}

void recursion3(int i, int j)
{
    if (j > N-i)
        return;
    printf("  ");
    recursion3(i, j+1);
}

void recursion1(int i)
{
    recursion3(i, 1);
    recursion2(i, 1);
    if (i >= N)
        return;
    recursion1(i+1);
    recursion3(i, 1);
    recursion2(i, 1);
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(1);
    return 0;
}
```

28.

```
        1
      2   2
    3   3   3
  4   4   4   4
5   5   5   5   5
  4   4   4   4
```

```
    3   3   3
      2   2
        1
```

```
#include<stdio.h>
int N;

void iteration()
{
    int i, j;
    for (i=1; i<=N; i++) {
        for (j=1; j<=N-i; j++)
            printf("  ");
        for (j=1; j<=i; j++)
            printf("%4d", i);
        printf("\n");
    }
    for (i=N-1; i>=1; i--) {
        for (j=1; j<=N-i; j++)
            printf("  ");
        for (j=1; j<=i; j++)
            printf("%4d", i);
        printf("\n");
    }
}

void recursion2(int i, int j)
{
    if (j > i) {
        printf("\n");
        return;
    }
    printf("%4d", i);
    recursion2(i, j+1);
}

void recursion3(int i, int j)
{
    if (j > N-i)
        return;
    printf("  ");
    recursion3(i, j+1);
}

void recursion1(int i)
{
    recursion3(i, 1);
    recursion2(i, 1);
    if (i >= N)
        return;
```

```
    recursion1(i+1);
    recursion3(i, 1);
    recursion2(i, 1);
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(1);
    return 0;
}
```

29.

```
1
2   2
3   3   3
4   4   4   4
5   5   5   5   5
4   4   4   4
3   3   3
2   2
1
```

```
#include<stdio.h>
int N;

void iteration()
{
    int i, j;
    for (i=1; i<=N; i++) {
        for (j=1; j<=i; j++)
            printf("%4d", i);
        printf("\n");
    }
    for (i=N-1; i>=1; i--) {
        for (j=1; j<=i; j++)
            printf("%4d", i);
        printf("\n");
    }
}

void recursion2(int i, int j)
{
    if (j > i) {
        printf("\n");
```

```
        return;
    }
    printf("%4d", i);
    recursion2(i, j+1);
}

void recursion1(int i)
{
    recursion2(i, 1);
    if (i >= N)
        return;
    recursion1(i+1);
    recursion2(i, 1);
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(1);
    return 0;
}
```

30.

```
                1
            2   2
        3   3   3
    4   4   4   4
5   5   5   5   5
    4   4   4   4
        3   3   3
            2   2
                1
```

```
#include<stdio.h>
int N;

void iteration()
{
    int i, j;
    for (i=1; i<=N; i++) {
        for (j=1; j<=N-i; j++)
            printf("    ");
        for (j=1; j<=i; j++)
            printf("%4d", i);
        printf("\n");
```

```
    }
    for (i=N-1; i>=1; i--) {
        for (j=1; j<=N-i; j++)
            printf("    ");
        for (j=1; j<=i; j++)
            printf("%4d", i);
        printf("\n");
    }
}

void recursion2(int i, int j)
{
    if (j > i) {
        printf("\n");
        return;
    }
    printf("%4d", i);
    recursion2(i, j+1);
}

void recursion3(int i, int j)
{
    if (j > N-i)
        return;
    printf("    ");
    recursion3(i, j+1);
}

void recursion1(int i)
{
    recursion3(i, 1);
    recursion2(i, 1);
    if (i >= N)
        return;
    recursion1(i+1);
    recursion3(i, 1);
    recursion2(i, 1);
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(1);
    return 0;
}
```

31.

```
               1
            1   2   1
        1   2   3   2   1
    1   2   3   4   3   2   1
1   2   3   4   5   4   3   2   1
```

```
#include<stdio.h>
int N;

void iteration()
{
    int i, j;
    for (i=1; i<=N; i++) {
        for (j=1; j<=N-i; j++)
            printf("    ");
        for (j=1; j<=i; j++)
            printf("%4d", j);
        for (j=i-1; j>=1; j--)
            printf("%4d", j);
        printf("\n");
    }
}

void recursion2(int i, int j)
{
    printf("%4d", j);
    if (j >= i) {
        return;
    }
    recursion2(i, j+1);
    printf("%4d", j);
}

void recursion3(int i, int j)
{
    if (j > N-i)
        return;
    printf("    ");
    recursion3(i, j+1);
}

void recursion1(int i)
{
    recursion3(i, 1);
    recursion2(i, 1);
    printf("\n");
    if (i >= N)
        return;
```

```
    recursion1(i+1);
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(1);
    return 0;
}
```

32.

```
                1
            1   2   1
        1   2   3   2   1
    1   2   3   4   3   2   1
1   2   3   4   5   4   3   2   1
    1   2   3   4   3   2   1
        1   2   3   2   1
            1   2   1
                1
```

```
#include<stdio.h>
int N;

void iteration()
{
    int i, j;
    for (i=1; i<=N; i++) {
        for (j=1; j<=N-i; j++)
            printf("    ");
        for (j=1; j<=i; j++)
            printf("%4d", j);
        for (j=i-1; j>=1; j--)
            printf("%4d", j);
        printf("\n");
    }
    for (i=N-1; i>=1; i--) {
        for (j=1; j<=N-i; j++)
            printf("    ");
        for (j=1; j<=i; j++)
            printf("%4d", j);
        for (j=i-1; j>=1; j--)
            printf("%4d", j);
        printf("\n");
    }
```

```
}

void recursion2(int i, int j)
{
    printf("%4d", j);
    if (j >= i) {
        return;
    }
    recursion2(i, j+1);
    printf("%4d", j);
}

void recursion3(int i, int j)
{
    if (j > N-i)
        return;
    printf("    ");
    recursion3(i, j+1);
}

void recursion1(int i)
{
    recursion3(i, 1);
    recursion2(i, 1);
    printf("\n");
    if (i >= N)
        return;
    recursion1(i+1);
    recursion3(i, 1);
    recursion2(i, 1);
    printf("\n");
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(1);
    return 0;
}
```

33.

```
                    1
                2   3   2
            3   4   5   4   3
        4   5   6   7   6   5   4
    5   6   7   8   9   8   7   6   5
```

```
    6   7   8   9  10  11  10   9   8   7   6
7   8   9  10  11  12  13  12  11  10   9   8   7
    6   7   8   9  10  11  10   9   8   7   6
        5   6   7   8   9   8   7   6   5
            4   5   6   7   6   5   4
                3   4   5   4   3
                    2   3   2
                        1
```

```
#include<stdio.h>
int N;

void iteration()
{
    int i, j;
    for (i=1; i<=N; i++) {
        for (j=1; j<=N-i; j++)
            printf("    ");
        for (j=1; j<=i; j++)
            printf("%4d", i+j-1);
        for (j=i-1; j>=1; j--)
            printf("%4d", i+j-1);
        printf("\n");
    }
    for (i=N-1; i>=1; i--) {
        for (j=1; j<=N-i; j++)
            printf("    ");
        for (j=1; j<=i; j++)
            printf("%4d", i+j-1);
        for (j=i-1; j>=1; j--)
            printf("%4d", i+j-1);
        printf("\n");
    }
}

void recursion2(int i, int j)
{
    printf("%4d", i+j-1);
    if (j >= i) {
        return;
    }
    recursion2(i, j+1);
    printf("%4d", i+j-1);
}

void recursion3(int i, int j)
{
    if (j > N-i)
        return;
    printf("    ");
```

```
    recursion3(i, j+1);
}

void recursion1(int i)
{
    recursion3(i, 1);
    recursion2(i, 1);
    printf("\n");
    if (i >= N)
        return;
    recursion1(i+1);
    recursion3(i, 1);
    recursion2(i, 1);
    printf("\n");
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(1);
    return 0;
}
```

34.

```
                    1
                2   3   2
            3   4   5   4   3
        4   5   6   7   6   5   4
    5   6   7   8   9   8   7   6   5
6   7   8   9   0   1   0   9   8   7   6
7   8   9   0   1   2   3   2   1   0   9   8   7
6   7   8   9   0   1   0   9   8   7   6
    5   6   7   8   9   8   7   6   5
        4   5   6   7   6   5   4
            3   4   5   4   3
                2   3   2
                    1
```

```
#include<stdio.h>
int N;

void iteration()
{
    int i, j;
    for (i=1; i<=N; i++) {
```

```
        for (j=1; j<=N-i; j++)
            printf("    ");
        for (j=1; j<=i; j++)
            printf("%4d", (i+j-1)%10);
        for (j=i-1; j>=1; j--)
            printf("%4d", (i+j-1)%10);
        printf("\n");
    }
    for (i=N-1; i>=1; i--) {
        for (j=1; j<=N-i; j++)
            printf("    ");
        for (j=1; j<=i; j++)
            printf("%4d", (i+j-1)%10);
        for (j=i-1; j>=1; j--)
            printf("%4d", (i+j-1)%10);
        printf("\n");
    }
}

void recursion2(int i, int j)
{
    printf("%4d", (i+j-1)%10);
    if (j >= i) {
        return;
    }
    recursion2(i, j+1);
    printf("%4d", (i+j-1)%10);
}

void recursion3(int i, int j)
{
    if (j > N-i)
        return;
    printf("    ");
    recursion3(i, j+1);
}

void recursion1(int i)
{
    recursion3(i, 1);
    recursion2(i, 1);
    printf("\n");
    if (i >= N)
        return;
    recursion1(i+1);
    recursion3(i, 1);
    recursion2(i, 1);
    printf("\n");
}

int main()
```

```
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    recursion1(1);
    return 0;
}
```

35.

```
   1
   2   3
   4   5   6
   7   8   9  10
  11  12  13  14  15
```

```
#include<stdio.h>
int N;
int k;

void iteration()
{
    int i, j;
    k = 1;
    for (i=1; i<=N; i++) {
        for (j=1; j<=i; j++)
            printf("%4d", k++);
        printf("\n");
    }
}

void recursion2(int i, int j)
{
    printf("%4d", k++);
    if (j >= i) {
        return;
    }
    recursion2(i, j+1);
}

void recursion1(int i)
{
    recursion2(i, 1);
    printf("\n");
    if (i >= N)
        return;
    recursion1(i+1);
}
```

```
int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    k = 1;
    recursion1(1);
    return 0;
}
```

36.

```
   1
   2   1
   3   3   1
   4   6   4   1
   5  10  10   5   1
   6  15  20  15   6   1
```

```
#include<stdio.h>
int N;
int k;

void iteration()
{
    int i, j;
    for (i=1; i<=N; i++) {
        k = 1;
        for (j=1; j<=i; j++) {
            k = k*(i-j+1)/j;
            printf("%4d", k);
        }
        printf("\n");
    }
}

void recursion2(int i, int j)
{
    k = k*(i-j+1)/j;
    printf("%4d", k);
    if (j >= i) {
        return;
    }
    recursion2(i, j+1);
}

void recursion1(int i)
```

```
{
    k = 1;
    recursion2(i, 1);
    printf("\n");
    if (i >= N)
        return;
    recursion1(i+1);
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    k = 1;
    recursion1(1);
    return 0;
}
```

37.

```
   1
   1   1
   1   2   1
   1   3   3   1
   1   4   6   4   1
   1   5  10  10   5   1
   1   6  15  20  15   6   1
```

```
#include<stdio.h>
int N;
int k;

void iteration()
{
    int i, j;
    for (i=1; i<=N; i++) {
        k = 1;
        for (j=1; j<=i; j++) {
            printf("%4d", k);
            k = k*(i-j)/j;
        }
        printf("\n");
    }
}

void recursion2(int i, int j)
{
```

```
    printf("%4d", k);
    k = k*(i-j)/j;
    if (j >= i) {
        return;
    }
    recursion2(i, j+1);
}

void recursion1(int i)
{
    k = 1;
    recursion2(i, 1);
    printf("\n");
    if (i >= N)
        return;
    recursion1(i+1);
}

int main()
{
    scanf("%d", &N);
    printf("Using Iteration: \n");
    iteration();
    printf("\nUsing Recursion: \n");
    k = 1;
    recursion1(1);
    return 0;
}
```